

1. 3번	2. 3번	3. 1번	4. 4번	5. 1번
6. 2번	7. 3번	8. 4번	9. 1번	10. 2번
11. 2번	12. 4번	13. 5번	14. 5번	15. 1번
16. 1번	17. 2번	18. 5번	19. 1번	20. 3번
21. 1번	22. 2번	23. 4번	24. 2번	25. 4번
26. 4번	27. 4번	28. 5번	29. 2번	30. 2번

$$\begin{aligned}
 1. \quad k &= \int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 x} \, dx = \int_0^{\frac{\pi}{4}} \sqrt{\sec^2 x} \, dx \\
 &= \int_0^{\frac{\pi}{4}} \sec x \, dx = [\ln(\sec x + \tan x)]_0^{\frac{\pi}{4}} \\
 &= \ln(\sqrt{2} + 1) \\
 \therefore e^k - \frac{1}{2} &= e^{\ln(\sqrt{2} + 1)} - \frac{1}{2} = \sqrt{2} + \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad &\int_2^e \left(\frac{1}{x \ln x} \right) dx + [x \ln x - x]_2^e \\
 &= \int_{\ln 2}^1 \frac{1}{t} dt + [x \ln x - x]_2^e \quad (\because \ln x = t) \\
 &= [\ln t]_{\ln 2}^1 + (-2 \ln 2 + 2) \\
 &= -\ln \ln 2 + (-2 \ln 2 + 2) \\
 &= -\ln 4 \ln 2 + 2
 \end{aligned}$$

$$\begin{aligned}
 3. \quad &\int_{\ln 2}^{\ln 5} \frac{e^x}{2e^x - 1} \, dx = \frac{1}{2} [\ln(2e^x - 1)]_{\ln 2}^{\ln 5} \\
 &= \frac{1}{2} (\ln 9 - \ln 3) = \frac{1}{2} \ln 3
 \end{aligned}$$

$$\begin{aligned}
 4. \quad &\int_{-1}^1 3x^2(x^3 + 1)^{\frac{1}{2}} \, dx = \frac{2}{3} [(x^3 + 1)^{\frac{3}{2}}]_{-1}^1 \quad (\because \text{전체적분}) \\
 &= \frac{4\sqrt{2}}{3}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad &\int_0^{\pi/2} \sin 2x \cdot e^{2\sin^2 x} \, dx \\
 &= \int_0^{\pi/2} 2 \sin x \cos x \cdot e^{2\sin^2 x} \, dx \quad (\because 2\text{배각공식}) \\
 &= \frac{1}{2} [e^{2\sin^2 x}]_0^{\pi/2} \quad (\because \text{전체적분}) \\
 &= \frac{1}{2} (e^2 - 1)
 \end{aligned}$$

$$\begin{aligned}
 6. \quad &\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} \, dx = \int_2^4 16 - t^2 \, dt \quad (\because \sqrt{16 - x^2} = t) \\
 &= \left[16t - \frac{1}{3} t^3 \right]_2^4 = \frac{40}{3}
 \end{aligned}$$

[다른 풀이1]

$$\begin{aligned}
 &\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} \, dx = \int_0^{\frac{\pi}{3}} 64 \sin^3 t \, dt \quad (x = 4 \sin t) \\
 &= \int_0^{\frac{\pi}{3}} 64 \sin t (1 - \cos^2 t) \, dt \\
 &= \left[-64 \cos t + \frac{64}{3} \cos^3 t \right]_0^{\frac{\pi}{3}} \\
 &= \frac{40}{3}
 \end{aligned}$$

[다른 풀이2]

$$\begin{aligned}
 &\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} \, dx = \int_0^{\frac{\pi}{3}} 64 \sin^3 t \, dt \text{에서} \\
 &\text{삼각함수 3배각 공식 } \sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha \text{를 활용하면} \\
 &\int_0^{\frac{\pi}{3}} 64 \sin^3 t \, dt = \int_0^{\frac{\pi}{3}} 48 \sin t - 16 \sin 3t \, dt \\
 &= \left[-48 \cos t + \frac{16}{3} \cos 3t \right]_0^{\frac{\pi}{3}} \\
 &= \frac{40}{3}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad &\int_{\frac{1}{2}}^1 \frac{x}{\sqrt{1 - (2x^2)^2}} \, dx = \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{1 - t^2}} \times \frac{1}{4} dt \quad (\because 2x^2 = t) \\
 &= \frac{1}{4} [\sin^{-1} t]_{\frac{1}{2}}^1 = \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 8. \quad &\int_{-2}^2 \sqrt{4 - x^2} \, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 \cos^2 \theta \, d\theta \quad (\because x = 2 \sin \theta) \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2(1 + \cos 2\theta) \, d\theta = [2\theta + \sin 2\theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2\pi
 \end{aligned}$$

[참고]

$y = \sqrt{4 - x^2}$ 은 반지름이 2인 반원이므로 $\int_{-2}^2 \sqrt{4 - x^2} \, dx$ 은 반원의 넓이를 이용하여 $\pi \times 2^2 \times \frac{1}{2} = 2\pi$ 로 풀어도 좋다.

$$\begin{aligned}
 9. \quad &\int_0^{1/2} \frac{x^2}{\sqrt{1 - x^2}} \, dx = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\cos \theta} \cos \theta \, d\theta \quad (\because x = \sin \theta) \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta \, d\theta = \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}} \\
 &= \frac{1}{2} \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 10. \quad &\int \frac{1}{\sqrt{x^2 - 16}} \, dx = \cosh^{-1} \frac{x}{4} + C \\
 &= \ln \left(\frac{x}{4} + \sqrt{\frac{x^2}{16} - 1} \right) + C = \ln \left(\frac{x + \sqrt{x^2 - 16}}{4} \right) + C
 \end{aligned}$$

[다른 풀이]

$$\begin{aligned} \int \frac{1}{\sqrt{x^2-16}} dx &= \int \frac{1}{4 \tan t} 4 \sec t \tan t dt \quad (\because x = 4 \sec t) \\ &= \int 4 \sec t dt \\ &= \ln(\sec t + \tan t) + C \\ \text{이때, } x &= 4 \sec t \text{ 이므로 } 1 + \tan^2 t = \sec^2 t \text{ 에서} \\ \tan t &= \frac{\sqrt{x^2-16}}{4} \text{ 이다.} \end{aligned}$$

$$\text{즉, } \ln(\sec t + \tan t) + C = \ln\left(\frac{x + \sqrt{x^2-16}}{4}\right) + C$$

$$\begin{aligned} 11. \int_1^2 \frac{1}{x^2 \sqrt{x^2+4}} dx &= \int_{\tan^{-1} \frac{1}{2}}^{\frac{\pi}{4}} \frac{\cos \theta}{4 \sin^2 \theta} d\theta \quad (\because x = 2 \tan \theta) \\ &= \int_{\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{2}}} \frac{1}{4 t^2} dt = -\left[\frac{1}{4 t}\right]_{\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{2}}} = \frac{\sqrt{5}-\sqrt{2}}{4} \end{aligned}$$

$$\begin{aligned} 12. \int_1^2 \frac{(\ln x)^2}{2} dx &= \frac{1}{2} \left[x(\ln x)^2 - \int 2 \ln x dx \right]_1^2 \quad (\because \text{부분적분}) \\ &= \frac{1}{2} [x(\ln x)^2 - 2(x \ln x - x)]_1^2 = (\ln 2)^2 - 2 \ln 2 + 1 \end{aligned}$$

$$\begin{aligned} 13. \int_0^{\frac{1}{2}} \cos^{-1} x dx &= \left[x \cos^{-1} x - \int \frac{-x}{\sqrt{1-x^2}} dx \right]_0^{\frac{1}{2}} \quad (\because \text{부분적분}) \\ &= [x \cos^{-1} x - \sqrt{1-x^2}]_0^{\frac{1}{2}} = \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \end{aligned}$$

[다른 풀이]

$\cos^{-1} x = t$ 로 치환하면 $\cos t = x$ 이고,

$$\begin{aligned} \int_0^{\frac{1}{2}} \cos^{-1} x dx &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} -t \sin t dt = -\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} t \sin t dt \\ &= [-t \cos t + \sin t]_{\frac{\pi}{3}}^{\frac{\pi}{2}} = \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \quad (\because \text{빠른 부분적분}) \end{aligned}$$

$$\begin{aligned} 14. \int_0^{\frac{1}{2}} \tan^{-1}(2x) dx &= x \tan^{-1}(2x) - \int \frac{2x}{1+4x^2} dx \quad (\because \text{부분적분}) \\ &= \left[x \tan^{-1}(2x) - \frac{1}{4} \ln(1+4x^2) \right]_0^{\frac{1}{2}} = \frac{\pi}{8} - \frac{1}{4} \ln 2 \end{aligned}$$

[다른 풀이]

$\tan^{-1}(2x) = t$ 로 치환하면 $x = \frac{1}{2} \tan t$ 이고

$$\begin{aligned} \int_0^{\frac{1}{2}} \tan^{-1}(2x) dx &= \int_0^{\frac{\pi}{4}} \frac{1}{2} t \sec^2 t dt \\ &= \frac{1}{2} [t \tan t + \ln \cos t]_0^{\frac{\pi}{4}} = \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$

$$\begin{aligned} 15. \int_0^1 (2x-1)e^x dx &= e^x(2x-1) - \int 2e^x dx \quad (\because \text{부분적분}) \\ &= [e^x(2x-1) - 2e^x]_0^1 = -e + 3 \end{aligned}$$

$$\begin{aligned} 16. \int_0^1 x^3 e^{x^2} dx &= \frac{1}{2} \int_0^1 t e^t dt \quad (\because x^2 = t) \\ &= \frac{1}{2} [te^t - \int e^t dt]_0^1 \quad (\because \text{부분적분}) \\ &= \frac{1}{2} [te^t - e^t]_0^1 = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} 17. \int_{\pi}^0 2e^x \cos x dx &= 2 \left[\frac{e^x}{2} (\cos x + \sin x) \right]_{\pi}^0 \quad (\because \text{오일러 적분}) \\ &= 1 + e^{\pi} \end{aligned}$$

[다른 풀이]

$$\begin{aligned} \int_{\pi}^0 e^x \cos x dx &= [e^x \cos x]_{\pi}^0 + \int_{\pi}^0 e^x \sin x dx \quad (\because \text{부분적분}) \\ &= 1 + e^{\pi} + [e^x \sin x]_{\pi}^0 - \int_{\pi}^0 e^x \cos x dx \quad (\because \text{부분적분}) \\ &= 1 + e^{\pi} - \int_{\pi}^0 e^x \cos x dx \text{ 에서} \\ \int_{\pi}^0 e^x \cos x dx &= \frac{e^{\pi} + 1}{2} \text{ 이다.} \end{aligned}$$

$$\begin{aligned} 18. \int_0^1 \frac{5x+7}{x^2+3x+2} dx &= \int_0^1 \frac{5x+7}{(x+1)(x+2)} dx \\ &= \int_0^1 \frac{2}{x+1} + \frac{3}{x+2} dx \\ &= [2 \ln(x+1) + 3 \ln(x+2)]_0^1 \\ &= 2 \ln 2 + 3 \ln 3 - 3 \ln 2 \\ &= 3 \ln 3 - \ln 2 \end{aligned}$$

$$\begin{aligned} 19. \int_0^1 x \sqrt{1-x} dx &= \int_0^1 2t^2 - 2t^4 dt \quad (\because \sqrt{1-x} = t) \\ &= \left[\frac{2}{3} t^3 - \frac{2}{5} t^5 \right]_0^1 = \frac{4}{15} \end{aligned}$$

$$\begin{aligned} 20. \int_0^{\pi^2} \cos \sqrt{t} dt &= \int_0^{\pi} 2x \cos x dx \quad (\because \sqrt{t} = x) \\ &= [2x \sin x - \int 2 \sin x dx]_0^{\pi} \quad (\because \text{부분적분}) \\ &= [2x \sin x + 2 \cos x]_0^{\pi} = -4 \end{aligned}$$

$$\begin{aligned} 21. \int_0^1 \arctan \sqrt{x} dx &= \int_0^1 2t \tan^{-1} t dt \quad (\because \sqrt{x} = t) \\ &= \left[t^2 \tan^{-1} t - \int \frac{t^2}{1+t^2} dt \right]_0^1 \quad (\because \text{부분적분}) \\ &= \left[t^2 \tan^{-1} t - \int 1 - \frac{1}{1+t^2} dt \right]_0^1 \\ &= [t^2 \tan^{-1} t - (t - \tan^{-1} t)]_0^1 = \frac{\pi}{2} - 1 \end{aligned}$$

$$\begin{aligned} 22. \int_0^3 \frac{x}{\sqrt{x+1}} dx &= \int_1^2 \frac{t^2-1}{t} \times 2t dt \quad (\because \sqrt{x+1} = t) \\ &= 2 \left[\frac{1}{3} t^3 - t \right]_1^2 = \frac{8}{3} \end{aligned}$$

$$\begin{aligned} 23. \int_2^3 \frac{9x+2}{\sqrt{3x-5}} dx &= \int_1^2 \frac{3(t^2+5)+2}{t} \times \frac{2}{3} t dt \quad (\because \sqrt{3x-5} = t) \\ &= \left[\frac{2}{3} t^3 + \frac{34}{3} t \right]_1^2 = 16 \end{aligned}$$

$$\begin{aligned} 24. \int_0^1 \frac{x}{\sqrt{3-2x-x^2}} dx &= \int_0^1 \frac{x}{\sqrt{4-(x+1)^2}} dx \\ &= \int_1^2 \frac{t-1}{\sqrt{4-t^2}} dt \quad (\because x+1 = t) \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{2 \sin \theta - 1}{2 \cos \theta} 2 \cos \theta d\theta \quad (\because t = 2 \sin \theta) \\ &= [-2 \cos \theta - \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = -\frac{\pi}{3} + \sqrt{3} \end{aligned}$$

$$25. \int_0^{\ln \sqrt{3}} \frac{1}{e^x + e^{-x}} dx = \int_1^{\sqrt{3}} \frac{1}{t + \frac{1}{t}} \times \frac{1}{t} dt \quad (\because e^x = t)$$

$$= \int_1^{\sqrt{3}} \frac{1}{t^2 + 1} dt = [\tan^{-1} t]_1^{\sqrt{3}} = \frac{\pi}{12}$$

$$26. \int_0^{\frac{\pi}{2}} \frac{5}{4\sin x + 3\cos x} dx \quad \left(\because \tan \frac{x}{2} = t \right)$$

$$= \int_0^1 \frac{5}{4 \times \frac{2t}{1+t^2} + 3 \times \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{10}{8t+3-3t^2} dt = \int_0^1 \frac{10}{(3-t)(1+3t)} dt$$

$$= \int_0^1 \frac{1}{3-t} + \frac{3}{1+3t} dt = [-\ln(3-t) + \ln(1+3t)]_0^1 = \ln 6$$

[다른 풀이]

삼각함수의 합성을 활용하면

$$4\sin x + 3\cos x = 5\sin(x+\alpha) \quad \left(\tan \alpha = \frac{3}{4} \right) \text{이므로}$$

$$\int_0^{\frac{\pi}{2}} \frac{5}{4\sin x + 3\cos x} dx = \int_0^{\frac{\pi}{2}} \csc(x+\alpha) dx$$

$$= [-\ln|\csc(x+\alpha) + \cot(x+\alpha)|]_0^{\frac{\pi}{2}}$$

$$= -\ln|\sec \alpha - \tan \alpha| + \ln|\csc \alpha + \cot \alpha|$$

$$= -\ln\left(\frac{5}{4} - \frac{3}{4}\right) + \ln\left(\frac{5}{3} + \frac{4}{3}\right)$$

$$= \ln 2 + \ln 3 = \ln 6$$

[참고]

$$\csc\left(\frac{\pi}{2} + \alpha\right) = \frac{1}{\sin\left(\frac{\pi}{2} + \alpha\right)} = \frac{1}{\cos \alpha} = \sec \alpha$$

$$\cot\left(\frac{\pi}{2} + \alpha\right) = -\tan \alpha \text{이다.}$$

$$27. \int_0^{\frac{\pi}{2}} \frac{1}{2 + \cos 2\theta} d\theta \quad (\because \tan \theta = t)$$

$$= \int_0^{\infty} \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^{\infty} \frac{1}{t^2 + 3} dt$$

$$= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^{\infty} = \frac{\pi}{2\sqrt{3}}$$

[다른 풀이]

$$\int_0^{\frac{\pi}{2}} \frac{1}{2 + \cos 2\theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{2 + 2\cos^2 \theta - 1} d\theta \quad (\because \text{반각공식})$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1 + 2\cos^2 \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{\cos^2 \theta + \sin^2 \theta + 2\cos^2 \theta} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{3\cos^2 \theta + \sin^2 \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{\sec^2 \theta}{3 + \tan^2 \theta} d\theta$$

$$= \int_0^{\infty} \frac{1}{3 + t^2} dt \quad (\because \tan \theta = t)$$

$$= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^{\infty} = \frac{\pi}{2\sqrt{3}}$$

$$28. \int_0^{\frac{1}{2}} \sin^{-1} x dx + \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$= [x \sin^{-1} x]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$+ \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx \quad (\because \text{부분적분법})$$

$$= \frac{\pi}{12}$$

$$29. \int_0^1 \sqrt{1+x^2} dx = \int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta \quad (\because x = \tan \theta)$$

$$= \frac{1}{2} [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta)]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} [\sqrt{2} + \ln(\sqrt{2} + 1)]$$

$$30. \int_3^4 \frac{1}{x^2 - 4} dx = \left[\frac{1}{4} \ln \left(\frac{x-2}{x+2} \right) \right]_3^4$$

$$= \frac{1}{4} \ln \left(\frac{1}{3} \right) - \frac{1}{4} \ln \left(\frac{1}{5} \right) = \frac{1}{4} \ln \frac{5}{3}$$

장황수학

1. 4번	2. 3번	3. 3번	4. 2번	5. 4번
6. 3번	7. 1번	8. 4번	9. 1번	10. 3번

1. $\tan^{-1}x = \alpha$ 라 하면 $\tan\alpha = x$, $\cos\alpha = \frac{1}{\sqrt{1+x^2}}$ 이다.

$$\begin{aligned} & \int_0^1 \sin\left(\tan^{-1}x + \cos^{-1}\frac{1}{\sqrt{1+x^2}}\right) dx \\ &= \int_0^1 \sin(2\alpha) dx = 2 \int_0^1 \sin(\alpha) \cos(\alpha) dx \\ &= 2 \int_0^1 \frac{x}{\sqrt{1+x^2}} \times \frac{1}{\sqrt{1+x^2}} dx = 2 \int_0^1 \frac{x}{1+x^2} dx \\ &= [\ln(1+x^2)]_0^1 = \ln 2 \end{aligned}$$

2. $f(2) - f(0) = \int_0^2 f'(x) dx = \int_0^2 \frac{1}{\sqrt{x^2+4}} dx = \left[\sinh^{-1} \frac{x}{2} \right]_0^2$
 $= \sinh^{-1} 1 = \ln(1 + \sqrt{2})$
 즉, $f(2) = \ln(1 + \sqrt{2})$

3. $\int_{\frac{1}{2}}^1 (\cos^{-1}x)^2 dx = - \int_{\frac{\pi}{3}}^0 t^2 \sin t dt \quad (\because \cos^{-1}x = t)$
 $= - \left[-t^2 \cos t + 2t \sin t + 2 \cos t \right]_{\frac{\pi}{3}}^0 = \frac{\sqrt{3}\pi}{3} - 1 - \frac{\pi^2}{18}$

4. $\int_0^\pi e^x \sin^2 x dx = \int_0^\pi e^x \left(\frac{1 - \cos 2x}{2} \right) dx$
 $= \frac{1}{2} \int_0^\pi e^x - e^x \cos 2x dx$
 $= \frac{1}{2} [e^x]_0^\pi - \frac{1}{2} \left[\frac{e^x}{5} (\cos 2x + 2 \sin 2x) \right]_0^\pi$
 $= \frac{1}{2} (e^\pi - 1) - \frac{1}{10} (e^\pi - 1) = \frac{2}{5} e^\pi - \frac{2}{5}$

5. $\int_1^{e^{\frac{\pi}{2}}} \sin(\ln x) dx \quad (\because \ln x = t)$
 $= \int_0^{\frac{\pi}{2}} e^t \sin t dt = \left[\frac{e^t}{2} (\sin t - \cos t) \right]_0^{\frac{\pi}{2}} = \frac{e^{\frac{\pi}{2}}}{2} + \frac{1}{2}$

6. $\int_0^1 \left(2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} \right) dx = \left[x^2 \sin \frac{1}{x^2} \right]_0^1 = \sin 1$

[다른 풀이]

$$\begin{aligned} & \int_0^1 \left(2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} \right) dx \\ &= \int_0^1 2x \sin \frac{1}{x^2} dx - \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx \\ &= \left[x^2 \sin \frac{1}{x} \right]_0^1 + \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx - \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx \quad (\text{부분 적분}) \\ &= \left[x^2 \sin \frac{1}{x} \right]_0^1 = \sin 1 \end{aligned}$$

7. $\int \frac{15x^2}{(x^2+1)(x^2+4)} dx = \int \frac{-5}{x^2+1} + \frac{20}{x^2+4} dx$
 $= -5 \tan^{-1}x + 10 \tan^{-1} \frac{x}{2} + C$

8. $\int \frac{1+2x^2}{x(1+x^2)^2} dx = \int \frac{x^2+1+x^2}{x(1+x^2)^2} dx = \int \frac{x}{(1+x^2)^2} + \frac{1}{x(1+x^2)} dx$
 $= \frac{-\frac{1}{2}}{1+x^2} + \int \frac{1}{x} + \frac{-x}{1+x^2} dx = \frac{1}{-2(1+x^2)} + \ln x - \frac{1}{2} \ln(1+x^2) + C$

[다른 풀이]

$$\int \frac{1+2x^2}{x(1+x^2)^2} dx = \int \frac{1}{x} + \frac{ax+b}{1+x^2} + \frac{cx+d}{(1+x^2)^2} dx$$

이때, 분자를 비교해보면

$$1+2x^2 = (1+x^2)^2 + (ax+b)x(1+x^2) + (cx+d)x$$

$$\Leftrightarrow 1+a=0, b=0, a+c=0, b+d=0$$

이므로 $a=-1, b=0, c=1, d=0$ 이다.

$$\int \frac{1}{x} + \frac{-x}{1+x^2} + \frac{x}{(1+x^2)^2} dx = \ln x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2(1+x^2)} + C$$

9. $\int \frac{1}{x^3+1} dx = \int \frac{1}{(x+1)(x^2-x+1)} dx$
 $= \int \frac{\frac{1}{3}}{x+1} + \frac{-\frac{1}{3}x + \frac{2}{3}}{x^2-x+1} dx$
 $= \int \frac{\frac{1}{3}}{x+1} + \frac{-\frac{1}{6}(2x-1) + \frac{1}{2}}{x^2-x+1} dx$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \int \frac{\frac{1}{2}}{x^2-x+1} dx$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{2} \int \frac{1}{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}} dx$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{2} \times \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}} + C$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{\sqrt{3}} \tan^{-1} \frac{2x-1}{\sqrt{3}} + C$

10. $\int \frac{f(x)}{x^2(x+1)^3} dx = \int \frac{a_1}{x} + \frac{a_2}{x^2} + \frac{a_3}{x+1} + \frac{a_4}{(x+1)^2} + \frac{a_5}{(x+1)^3} dx$
 의 부정적분이 유리함수가 되려면 $a_1 = a_3 = 0$ 이다.

$$\text{또한 } \int \frac{A}{x^2} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} dx$$

$$= \int \frac{A(x+1)^3 + Bx^2(x+1) + Cx^2}{x^2(x+1)^3} dx \text{ 이므로}$$

$$f(x) = ax^2 + bx + c = A(x+1)^3 + Bx^2(x+1) + Cx^2 \text{ 이다.}$$

$$f(0) = 1 \text{ 이므로 } c = 1 \text{ 이다.}$$

$$\text{이때, 계수 비교를 하면 } A+B=0, 3A+B+C=a,$$

$$3A=b, A=c=1 \text{ 이다.}$$

$$A=1, B=-1, b=3, c=1 \text{ 이므로 } f(x) = ax^2 + 3x + 1 \text{ 이다.}$$

$$\text{즉, } f'(x) = 2ax + 3 \text{ 이므로 } f'(0) = 3 \text{ 이다.}$$

1. 4번	2. 4번	3. 1번	4. 1번	5. 2번
6. 4번	7. 4번	8. 1번	9. 18	10. 3번
11. 1번				

$$\begin{aligned}
 1. \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^4 x dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^2 x \cdot \csc^2 x dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 + \cot^2 x) \csc^2 x dx = \int_1^0 -(1+t^2) dt \quad (\because \cot x = t) \\
 &= -\left[t + \frac{1}{3}t^3\right]_1^0 = \frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 2. \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} [(\tan x)^{-2} \sec^4 x + \tan^3 x (\sec x)^{-1}] dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left((\tan x)^{-2} (1 + \tan^2 x) \sec^2 x + \frac{\sin^3 x \times \cos x}{\cos^3 x} \right) dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{\tan^2 x} + 1 \right) \sec^2 x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{\cos^2 x} - 1 \right) \sin x dx \\
 &= \int_1^{\sqrt{3}} \left(\frac{1}{t^2} + 1 \right) dt + \int_{\frac{1}{2}}^1 \left(\frac{1}{u^2} - 1 \right) du \quad (\because \tan x = t, \cos x = u) \\
 &= \left[-\frac{1}{t} + t \right]_1^{\sqrt{3}} + \left[-\frac{1}{u} - u \right]_{\frac{1}{2}}^1 \\
 &= \frac{2}{3}\sqrt{3} - \frac{3}{2}\sqrt{2} + \frac{5}{2}
 \end{aligned}$$

$$\begin{aligned}
 3. \int_{\frac{1}{2}}^1 \sqrt{\frac{1}{x} - 1} dx &= \int_{\frac{1}{2}}^1 \sqrt{\frac{1-x}{x}} dx = \int_{\frac{1}{2}}^1 \frac{\sqrt{1-x}}{\sqrt{x}} dx \\
 &= \int_{\frac{1}{\sqrt{2}}}^1 \frac{\sqrt{1-t^2}}{t} 2t dt \quad (\because \sqrt{x} = t) \\
 &= 2 \int_{\frac{1}{\sqrt{2}}}^1 \sqrt{1-t^2} dt = 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 \theta d\theta \quad (\because t = \sin \theta) \\
 &= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 + \cos 2\theta}{2} d\theta = \left[\theta + \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 4. \int_0^2 \tan^{-1} \frac{2-x}{1+2x} dx &= \int_0^2 \tan^{-1} 2 - \tan^{-1} x dx \\
 &= 2 \tan^{-1} 2 - \int_0^2 \tan^{-1} x dx \\
 &= 2 \tan^{-1} 2 - \left[x \tan^{-1} x - \int \frac{x}{1+x^2} dx \right]_0^2 \quad (\because \text{부분적분}) \\
 &= 2 \tan^{-1} 2 - \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^2 = \frac{1}{2} \ln 5
 \end{aligned}$$

$$\begin{aligned}
 5. \int_0^{2a} x \sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) dx \\
 &= \left[\frac{1}{2} x^2 \sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) \right]_0^{2a} + \frac{1}{4} \int_0^{2a} \frac{x^2}{\sqrt{4a^2-x^2}} dx \quad (\because \text{부분적분}) \\
 &= \frac{1}{4} \int_0^{2a} \frac{x^2}{\sqrt{4a^2-x^2}} dx = a^2 \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta \quad (\because x = 2a \sin \theta) \\
 &= \frac{a^2}{2} \int_0^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta
 \end{aligned}$$

$$= \frac{a^2}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4} a^2$$

[다른 풀이]

$$\sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) = \theta \text{ 라 하면 } \sin \theta = \frac{\sqrt{2a-x}}{2\sqrt{a}} \text{ 이고}$$

양변을 제곱하면 $4a \sin^2 \theta = 2a - x$

$$\Rightarrow x = 2a - 4a \sin^2 \theta = 2a - 4a(1 - \cos^2 \theta)$$

$$= 4a \cos^2 \theta - 2a = 2a \cos 2\theta \text{ 이므로}$$

$$\int_0^{2a} x \sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) dx$$

$$= \int_{\frac{\pi}{4}}^0 2a \cos 2\theta \cdot (-4a \sin 2\theta) d\theta$$

$$= 4a^2 \int_0^{\frac{\pi}{4}} \theta \sin 4\theta d\theta$$

$$= 4a^2 \left[-\frac{\theta}{4} \cos 4\theta + \frac{1}{16} \sin 4\theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} a^2 \text{ 이다.}$$

6. $(e^x f(x))' = e^x (f(x) + f'(x))$ 임을 이용하자.

$$\int_0^1 \frac{x^3 e^{x^2}}{(x^2+1)^2} dx = \int_0^1 \frac{x x^2 e^{x^2}}{(x^2+1)^2} dx \quad (\because x^2+1=t)$$

$$= \int_1^2 \frac{(t-1)e^{t-1}}{t^2} \frac{1}{2} dt$$

$$= \frac{1}{2e} \int_1^2 \frac{(t-1)e^t}{t^2} dt$$

$$= \frac{1}{2e} \int_1^2 e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt$$

$$= \frac{1}{2e} \left[e^t \frac{1}{t} \right]_1^2 \quad (\because \text{곱미분})$$

$$= \frac{1}{4} e - \frac{1}{2}$$

$$\begin{aligned}
 7. \int_1^\infty \frac{2x^3+3x^2+3}{x^3(x^2+1)} dx &= \int_1^\infty \frac{2x^3+3(x^2+1)}{x^3(x^2+1)} dx \\
 &= \int_1^\infty \frac{2}{x^2+1} + \frac{3}{x^3} dx = \left[2 \tan^{-1} x - \frac{3}{2x^2} \right]_1^\infty = \frac{\pi}{2} + \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 8. \int_0^1 \frac{9}{1+x^3} dx &= \int_0^1 \frac{9}{(x+1)(x^2-x+1)} dx \\
 &= \int_0^1 \left(\frac{3}{x+1} + \frac{-3x+6}{x^2-x+1} \right) dx \\
 &= [3 \ln(1+x)]_0^1 - \frac{3}{2} \int_0^1 \frac{(2x-1)-3}{x^2-x+1} dx \\
 &= 3 \ln 2 - \frac{3}{2} \int_0^1 \frac{2x-1}{x^2-x+1} dx + \frac{9}{2} \int_0^1 \frac{1}{x^2-x+1} dx \\
 &= 3 \ln 2 - \frac{3}{2} [\ln(x^2-x+1)]_0^1 + \frac{9}{2} \int_0^1 \frac{1}{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}} dx \\
 &= 3 \ln 2 + 3\sqrt{3} \left[\tan^{-1} \left(\frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1 \\
 &= \ln 8 + \sqrt{3} \pi
 \end{aligned}$$

$$\begin{aligned}
9. & \int_{1+\sqrt{3}}^{\infty} \left(\frac{12}{x-1} - \frac{12(x+2)}{x^2+x+1} \right) dx \\
&= \int_{1+\sqrt{3}}^{\infty} \left(\frac{12}{x-1} - \frac{6(2x+1)+18}{x^2+x+1} \right) dx \\
&= \int_{1+\sqrt{3}}^{\infty} \left(\frac{12}{x-1} - \frac{6(2x+1)}{x^2+x+1} - \frac{18}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \right) dx \\
&= \left[12 \ln |x-1| - 6 \ln |x^2+x+1| \right]_{1+\sqrt{3}}^{\infty} \\
&\quad - \left[18 \times \frac{2}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_{1+\sqrt{3}}^{\infty} \\
&= \left[6 \ln \frac{(x-1)^2}{x^2+x+1} - 12 \sqrt{3} \tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_{1+\sqrt{3}}^{\infty} \\
&= 6 \ln(2+\sqrt{3}) - 12 \sqrt{3} \left(\frac{\pi}{2} - \tan^{-1} \frac{3+2\sqrt{3}}{\sqrt{3}} \right) \\
&= 6 \ln(2+\sqrt{3}) - 12 \sqrt{3} \left(\frac{\pi}{2} - \tan^{-1}(2+\sqrt{3}) \right) \\
&= 6 \ln(2+\sqrt{3}) - 6 \sqrt{3} (\pi - 2 \tan^{-1}(2+\sqrt{3})) \\
&= 6 \ln(2+\sqrt{3}) - 6 \sqrt{3} \times \frac{\pi}{6} \quad (\because \text{아래부분 참고}) \\
&= -\sqrt{3} \pi + 6 \ln(2+\sqrt{3}) \\
&\text{즉, } a=3, b=6 \text{ 이므로 } ab=18
\end{aligned}$$

[참고]

$\tan^{-1}(2+\sqrt{3})=X$ 라 하면 $\tan X=2+\sqrt{3}$ 이다.

$$\Rightarrow \tan(\pi - 2 \tan^{-1}(2+\sqrt{3}))$$

$$= \tan(\pi - 2X) = -\tan 2X$$

$$= -\frac{2 \tan X}{1 - \tan^2 X} = -\frac{2(2+\sqrt{3})}{1 - (2+\sqrt{3})^2} = \frac{1}{\sqrt{3}}$$

$$\text{즉, } \pi - 2 \tan^{-1}(2+\sqrt{3}) = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$\Rightarrow \tan^{-1}(2+\sqrt{3}) = \frac{5}{12} \pi$$

$$\tan(15^\circ) = 2 - \sqrt{3}, \tan(75^\circ) = 2 + \sqrt{3}$$

$$\begin{aligned}
10. & I_7 + I_9 = \int_0^{\frac{\pi}{4}} \tan^7 x dx + \int_0^{\frac{\pi}{4}} \tan^9 x dx \\
&= \int_0^{\frac{\pi}{4}} \tan^7 x (1 + \tan^2 x) dx = \int_0^{\frac{\pi}{4}} \tan^7 x \sec^2 x dx \\
&= \frac{1}{8} [\tan^8 x]_0^{\frac{\pi}{4}} = \frac{1}{8}
\end{aligned}$$

[참고]

$$I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx \text{ 일 때, } I_{n+2} + I_n = \frac{1}{n+1}$$

$$I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx \text{ 일 때, } I_{n+2} + I_n = \frac{1}{n+1}$$

$$\begin{aligned}
11. & \int_0^1 \frac{\sec^2 x - \tan x}{e^x} dx \\
&= \int_0^1 e^{-x} \sec^2 x dx - \int_0^1 e^{-x} \tan x dx \\
&= [e^{-x} \tan x]_0^1 + \int_0^1 e^{-x} \tan x dx - \int_0^1 e^{-x} \tan x dx \quad (\because \text{부분적분}) \\
&= \tan(1) e^{-1}
\end{aligned}$$

[다른 풀이]

$$(e^{-x} \tan x)' = -e^{-x} \tan x + e^{-x} \sec^2 x \text{ 이므로}$$

$$\int_0^1 \frac{\sec^2 x - \tan x}{e^x} dx = [e^{-x} \tan x]_0^1 = \tan(1) e^{-1}$$